Data descriptor

vocabmod-se — a structured resource for Swedish health-data terminologies and cross-system mappings

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In brief

vocabmod-se links the codes used in Swedish health data to the international terminologies that research and cross-border exchange rely on. It holds twenty-three Swedish code systems, among them ICD-10-SE for diagnoses, KVÅ for procedures, TLV's registers for medicines and consumables, NPU and six hospitals' local catalogues for laboratory tests, and ICF and KSI for functioning and social services. From these systems run 110,101 mappings to SNOMED CT, LOINC, ATC, RxNorm, and other reference terminologies, and to the concepts of the OMOP common data model that most observational health research uses. A data engineer can load Swedish codes into an OMOP database beside OHDSI's standard vocabularies, and a researcher or an agency can look up what a Swedish code corresponds to elsewhere. Everything is plain text in a public repository, checked against one schema, and a companion command-line tool exports it as OMOP vocabulary tables or FHIR ConceptMaps. The national classifications are carried as codes and hierarchies only, and licensed terminologies are referenced by code alone.

A Swedish code counts as covered when one of its mappings leads to a standard OMOP concept, directly or through one "Maps to" step. By that rule the mappings cover 58,260 of the 71,146 Swedish codes in scope (81.9%); most of the codes left uncovered are medical consumables and specialized laboratory properties, and review found no suitable standard concept for most of them. Every mapping carries a confidence tier or the reason why none could be assigned. 19,073 mappings (17%) are in the high tier, which agreed with a blinded, rubric-based review in 90% of cases (95% CI 78–96) on a held-out sample, so these mappings are reasonable candidates for lower priority in expert review; 45% of the mappings could not be assessed. The blinded reviews were made by language-model agents working to a written rubric. No domain expert has reviewed the mappings, and a tier grades the evidence behind a mapping, not its correctness.

vocabmod-se is an independent project that I build and publish on my own, part-time. The first mappings originate in GMLF, a Swedish emergency-care research project, where they were made as an exploratory step in assessing the project's data against OMOP, not as a research output in their own right, and they covered only part of the material. Everything since is my own work, done independently of that project: the extended mappings, the review that lifted coverage to its present level, the schema, and the tool. I have relied extensively on AI coding agents for the mappings, the tool, and this report; work of this extent would not otherwise have been feasible for one person working part-time. Mapping Swedish clinical code systems to international ones such as SNOMED CT and OMOP is a matter of broad national interest, and I publish the material so that others doing such work can build on what was found and learned. It is released as open source, not as a proposed standard: it offers the material gathered so far and one way such work could be structured. I make no commitment to long-term maintenance, but researchers and professionals are welcome to open issues, send pull requests, or start discussions, and a dedicated effort would be welcome to take the work further.

The corpus, this report, and the tool are published at github.com/ludvigalden/vocabmod-se.

1. What the corpus holds

The corpus holds twenty-three Swedish code systems with 89,497 concepts, and the mappings that connect them to one another and to international terminologies. Table 1 groups the systems by clinical domain.

Domain	Systems	Concepts
Diagnoses	ICD-10 (WHO base), ICD-10-SE, ICD-O-3 morphology	39,303
Procedures	KVÅ surgical and medical, NCSP	15,650
Pharmaceuticals	TLV package numbers, product groups, consumables, dose forms; ATC Y-codes	19,117
Laboratory	NPU, six hospitals' local catalogues, Swedish unit spellings	11,213
Function and social services	ICF, KSI, SPKOD	3,858
Total		**89,497**

The total includes 356 concepts in four small registers (counties and municipalities, civil status, education levels, and triage levels), which the table does not show as rows. Every count in this report can be checked against the committed census (`_registry/census.kdl`).

Most concepts carry a code but no name: 19,928 carry at least one designation, and the other 69,569 (77.7%) carry none. For the national classifications — ICD-10-SE, KVÅ (KKÅ and KMÅ), the Swedish ICF, and KSI, which E-hälsomyndigheten has published since 1 June 2026, SPKOD from Socialstyrelsen, and the rest — that is a publication boundary rather than a source gap: their publishers grant no license to redistribute the designation catalogues, so the tree holds codes, hierarchy, and validity. Where terms do permit carriage, the designations are there: NPU's property strings verbatim under the NPU terms of use, SCB's county and municipality names under CC0, and RxNorm product names by permission of NLM. The English KVÅ designations are this project's own renderings of the Swedish codes.

The mappings sit in versioned collections, each from one source system to one target system. Of the 110,101 mappings, 44,450 (40%) target SNOMED CT and 37,945 (34%) target ATC. Most state the Swedish code as narrower than its target (82.1%), as when a medicine package maps to the ATC class that contains it; 15.8% state equivalence, and the rest partial overlap, a looser relation, or a broader source. Counted apart are 8,337

identifier bindings, which give a Swedish code an OMOP concept id without claiming any relationship of meaning.

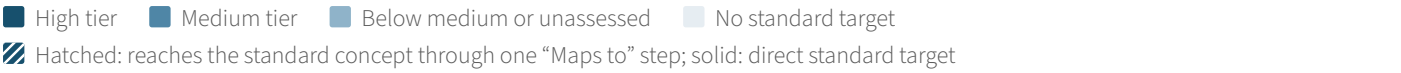
For a reader preparing Swedish data for European exchange, Table 2 sets the mappings, element by element, against the code systems the eHealth Network guidelines and HL7 Europe implementation guides name (non-binding): the Patient Summary Guideline (release 3.4), the ePrescription and eDispensation Guideline (release 3.1), and the HL7 Europe FHIR guides. Regulation (EU) 2025/327 names no code system itself; the coding of the European electronic health record exchange format is left to an implementing act due by March 2027, not adopted as of November 2026.

Data element	Named systems	Corpus collections	Mappings	Quality-scored
Diagnoses	ICD-10, SNOMED CT	ICD-10-SE → SNOMED CT, also as OMOP ids and via code truncation; ICD-O-3 → SNOMED CT	31,420	1,811
Rare diseases	Orphacode	ICD-10-SE → Orphanet	230	0
Procedures	SNOMED CT	KVÅ surgical and medical → SNOMED CT (7 to LOINC and RxNorm)	8,051	805
Laboratory results	LOINC, NPU, SNOMED CT	hospital catalogues → LOINC, NPU; NPU → LOINC, SNOMED CT	6,371	552
Units	UCUM	Swedish unit spellings → UCUM	14	0
Medication class	ATC	TLV package numbers → ATC; RxNorm → ATC	37,945	0
Active substance	EMA SMS	none	0	0
Dose form	EDQM	TLV dose forms → EDQM; SNOMED CT → EDQM	514	0
Route	EDQM	none	0	0
Functional status	ICF	KSI → ICF; ICF → SNOMED CT, LOINC	3,086	112
Vaccines	SNOMED CT, ATC	three TLV packages → ATC J07	3	0
Allergies	SNOMED CT, ATC	none	0	0
Pregnancy	SNOMED CT	none	0	0
Document types	LOINC	none	0	0

Census counts; an arrow reads source to target, and quality-scored counts the mappings that carry the rubric score of chapter 3, a narrower grade than the confidence tiers. ICD-10-SE extends ICD-10, so its codes are ICD-10 codes or children of one. The vaccine packages are counted in the medication-class row, and the 153 SNOMED CT mappings of ICD-10-SE chapter O (pregnancy, childbirth, and the puerperium) in the diagnosis row. Care plans are left out, because the guidance binds no code system to them. The mappings reach eight of the fourteen elements, and vaccines only through three packages. Quality scores are densest for procedures and laboratory results; the medication, dose-form, unit, and rare-disease rows carry none.

2. How far the mappings reach

Most uses of the mappings need a Swedish code to end at a concept the OMOP common data model treats as standard, because analyses and exports work with those concepts. A code therefore counts as covered only when a mapping or identifier binding leads to an OMOP-standard concept, directly or through one “Maps to” step. By that rule the mappings cover 58,260 of the 71,146 Swedish codes in scope, 81.9%; the figure shows the share per system and the confidence tier (chapter 3) of the covering mappings.



System	Covered	Codes	%	Coverage by tier
ICD-10-SE	26,414	26,446	99.9	
TLV package numbers	13,473	13,852	97.3	
NPU laboratory codes	4,272	9,878	43.2	
KKÅ procedures	5,174	6,899	75.0	
TLV consumables and aids	1,011	4,339	23.3	
KMÅ procedures	2,780	3,497	79.5	
KSI social services	1,604	2,074	77.3	
ICF function	1,348	1,632	82.6	
Hospital laboratory catalogues	1,315	1,320	99.6	
TLV product groups	199	388	51.3	
ATC Y-codes	164	298	55.0	
SPKOD	150	152	98.7	
Total, 12 systems	**57,904**	**70,775**	**81.8**	

Coverage and confidence per Swedish system under the target-standard rule: bar length is the covered share, shade the confidence tier of the covering mappings (chapter 3), and the hatch marks coverage that reaches a standard concept only through one “Maps to” step. Not shown: four systems below fifty codes and the Swedish geography codes, custom concepts by nature: Triage levels (18 of 29); Swedish units (14 of 15); Education (6 of 9); Civil status (7 of 7); Counties and municipalities (311 of 311), together 356 of 371 codes covered. TLV dose forms (240 codes) are out of scope, reported as attributes rather than gaps.

Two registers hold most of the codes the rule does not reach. TLV’s register of consumables and medical aids leaves 3,328 of its 4,339 codes uncovered, mostly products and devices for which no standard vocabulary offers a target. NPU leaves 5,606 of its 9,878 laboratory codes uncovered, among them blood-group antibody concentrations, allergen-specific IgE, and HLA typing, which no standard vocabulary names. These gaps have been examined, not only counted. The review described in chapter 3 worked through all 22,682 codes that were uncovered before it and covered 9,796 of them.

Of the 12,886 that remain uncovered, reviewers found no suitable standard concept among the candidates for 10,625; for 2,174 the two blind review passes chose different targets, 75 lacked the information needed to judge, and 12 have no name to search with.

Dose forms lie outside the rule. TLV's 240 dose forms are mapped to EDQM standard terms as attributes of a package, not as codes to be covered, so they are left out of the 71,146.

3. How the mappings were made and graded

The mappings come from three sources, and they differ in how much evidence stands behind each one. That is why every mapping is graded.

The oldest mappings were imported: publisher crosswalks and register material, and the early mappings from GMLF, made there for an initial assessment of the project's laboratory data from six hospitals and of how that data could be mapped to OMOP. A program converted this material into the published form. It is deterministic, so repeated runs over the same input give the same tree byte for byte, and a source row it cannot interpret stops the run rather than being dropped. Labels from the sources were normalized through four translation tables committed with the registry, and the six hospital catalogues were merged under a published rule into 1,320 concepts, with 54 local spellings kept as variants. Because its sources cannot be redistributed, the program is not published: a reader can re-validate the tree and re-derive every statistic, but not re-run the import.

The second source is composition from what the corpus already holds: 311 county and municipality codes bound to their OMOP concept ids, 1,170 KSI codes mapped onward through ICF mappings that end at standard concepts, and 451 ICD-10-SE codes mapped to ICD-10-CM parents by shortening the code, which caps them at medium confidence.

The third source is a review of every code that was still uncovered, and it added 9,889 mappings. For each code, BioLORD-2023-M, a multilingual biomedical language model, proposed candidate concepts from OMOP's standard vocabularies; on a held-out sample the accepted concept was among its top ten for 59.0% of the codes where one existed. Language-model agents then judged the candidates against a written rubric without seeing the model's ranking, in two independent blind passes. 8,987 mappings were committed because both passes chose the same concept. For 845 codes the two passes chose different concepts that were both defensible; a third blind pass, shown both without knowing which pass chose which, picked one for 788 codes and kept both as alternatives for 57, adding 902 mappings. Two such passes agree about 81% of the time, which is the ceiling on how precisely this kind of review can be measured.

Every mapping carries a confidence tier, or one of five named reasons why it cannot be graded. A tier is decided by three facts: whether the target is OMOP-standard, where BioLORD ranks the target among its candidates, and how high it scores. A standard target ranked first above a fixed score is high, ranked first below it or second or third is medium, and ranked lower is low. The rule was fit on 588 reviewed cases and tested on a held-out sample. All 9,889 review mappings are high because blind review accepted them,

the 902 tie-break mappings included; the held-out agreement below was measured on the rule, not on them.

Tier	Mappings	Share	Held-out agreement with blinded review
High	19,073	17%	90% (95% CI 78–96)
Medium	8,837	8%	71.1% (95% CI 55–83)
Low	32,996	30%	5.6%
Unassessable	49,195	45%	—

The high tier agreed with the blinded, rubric-based review in 90% of cases (95% CI 78–96) on a held-out sample, so its mappings are reasonable candidates for lower priority in expert review. A low tier means that nothing independent corroborates the target, not that the target is wrong. Most unassessable mappings have a target outside OMOP's vocabularies or a source that is not a Swedish code, as in the RxNorm-to-ATC crosswalks. Precision above the roughly 81% two-pass ceiling cannot be told apart from reviewer disagreement, and no domain expert has reviewed the released mappings.

An older and narrower grade sits beside the tiers. A published rubric reads the comment that records each mapping's basis and derives a quality score, and an effect where the comment states that the mapping loses part of the source's meaning. Stated losses rank first, so a mapping that admits a loss never gets a full score. 3,311 mappings (3.0%) carry a score: 1,937 at 100 (a verified or exact basis), 679 at 60 (a step up the hierarchy), and 695 at 40 (a stated approximation or lost detail). A mapping without a score has not been assessed by this rubric, which says nothing about whether it is wrong.

4. Using the corpus

The corpus can be read without running any software. It is plain text under `vocab/`, one directory per code system, and each system's `definitions.kdl` names its release and its mapping collections with their source and target systems. The tree can be re-validated against the committed schema with the kdlith document processor, whose revision `sources.kdl` pins. To cite the corpus, cite the commit you used rather than a branch; `CITATION.cff` carries the citation metadata.

A mapping record says what it connects (a `source` code and a `target`), what it claims (its `relationship`), whether it is the first choice or a fallback (`preference`, `primary` or `alternative`), and why it exists, in its comment. The comment is the whole basis of the mapping and the only input to its quality score, so it is also where a reader who disagrees can point to the fault. An `omop-binding` record is weaker: it gives a code an OMOP concept id without claiming a relationship of meaning, and should not be read as a mapping.

The companion command-line tool, `vocabmod-se`, turns the corpus into the formats other systems load. `omop-export` writes the seven OMOP vocabulary tables the corpus can fill, ready to load beside OHDSI's Athena vocabularies: one CONCEPT row for every concept without an official OMOP id (89,171 rows), with ids from 2,000,000,000 upwards derived from each code by a stable hash. `fhir-export` writes one FHIR R5 ConceptMap per mapping collection, its targets referenced by code, with display text only where the corpus carries its own designation. `search` looks up names and codes, and `import` turns a locally held source, such as a licensed designation file, into a layer that never enters the repository.

Two further layers, for a reader moving Swedish diagnosis data towards ICD-11, are read by the tool but not published in the repository. Both are built from WHO's ICD-10↔ICD-11 mapping tables for the ICD-11 MMS 2026-01 release and joined on WHO's version-stable Foundation identifiers. WHO's terms reserve crosswalks to ICD-10 to a separate agreement, so these layers are held outside the committed corpus and merged into a local reading with the tool's `--imports` option.

The ICD-10 layer holds 28,227 paths over 12,597 ICD-10 entities and reaches every one of ICD-10's 11,243 categories. The paths are evidence for a reviewer, never mapping targets in their own right. The Swedish layer holds 56,262 paths for the 25,158 rows of the Swedish 2026 diagnosis extract that have no mapping of their own. A Swedish extension code never matches an ICD-10 code exactly, but 25,013 of the rows (99.4%) reach ICD-11 through their ICD-10 parent. The 145 rows without a path are national extensions with no ICD-10 parent, and they are left without one.

5. Licensing and publication

What may be published decides what the corpus carries, and the repository's LICENSE file states the terms and attribution for each source. Published are the codes, hierarchies, and validity of every system; the mappings with their comments and grades; and the project's own work — schema, tool, registries, the English KVÅ renderings, and the laboratory mapping material. The designation catalogues of the Swedish national classifications are not redistributed: those systems carry codes, hierarchy, and validity. Where a review judged a mapping, its rationale may quote the single title it judged from the Swedish national classifications or TLV's published lists, with the source named, under the quotation right of the Swedish Copyright Act; RETTS triage titles are not quoted.

Each omission rests on a publicly stated term. WHO's ICD-10, 2016 edition, carries an all-rights-reserved imprint and the 2019 edition is offered under CC BY-ND 3.0 IGO; neither licenses redistribution of designation text, and the same license reserves crosswalks between other classifications and ICD-10 to a separate written agreement, the clause under which the ICD-10 crosswalk collections (21,920 mappings) and the ICD-11 layers of chapter 4 are held outside the corpus. The Swedish classifications are protected by copyright with quotation permitted, and the publishers' terms grant no license to redistribute their designation catalogues, so KVÅ, KSI, SPKOD, ICD-10-SE, and the Swedish ICF are carried as codes, with single titles quoted only as review evidence. The Nordic procedure classification has no public reuse license, and the triage point names belong to a commercially licensed product: both code-only. The pharmaceutical registers are published for reuse under lag (2022:818), but without an explicit license text, so they are carried as codes, with single titles quoted only as review evidence. The designations the corpus does carry are carried under their own terms: NPU verbatim under the NPU terms of use with attribution, SCB under CC0, and RxNorm names as U.S. public domain with the NLM acknowledgement. SNOMED CT, LOINC, MedDRA, and UCUM are referenced by code only, under the reader's own license with each owner; EDQM standard terms are referenced by code and OMOP concept identifiers by id alone; Orphanet codes are reused under CC BY 4.0 with the attribution LICENSE gives.

The corpus is versioned by repository release, and every mapping collection names its own version, 2026.1 throughout this release.

6. Limitations

The largest limitation is that no domain expert has reviewed the mappings. The confidence tiers of chapter 3 grade the evidence behind each mapping, 45% of the mappings could not be graded at all, and only 3,311 (3.0%) carry the rubric's quality score. Before a mapping is relied on, it should be checked against its comment and, for any consequential use, by someone qualified to judge it.

The corpus reflects the releases it imported. KVÅ, ICD-10-SE, and the TLV registers are snapshots, so a code added after its snapshot is missing. Designations limit retrieval too: the candidate search works from a code's name, so a code without one has nothing to search with, and 2,147 mappings are unassessable for that reason.

Six data elements of Table 2 lie outside the corpus, in the table's order. Medication reaches ATC classes and dose forms but no active substances (EMA SMS) and no route of administration. Immunization reaches only three ATC J07 packages, since most vaccines sit outside the TLV benefit registers. Allergies, pregnancy, and document types have no collection; closing them needs a Swedish source system the corpus does not carry.

Two smaller points concern particular records. The 103 laboratory mappings that keep their legacy ten-digit NPU identifiers have no reliable route to the current spelling, so a user resolving them against a current NPU release has to translate them. And the project's reserved OMOP custom-concept ranges are a convention for sharing between users of this corpus, never OHDSI content.

The corpus is meant for looking up what Swedish health-data codes correspond to in reference terminologies, for seeing where a mapping came from and what it claims, and for building OMOP vocabularies to use beside Athena, reproducibly from a pinned commit. It is not a terminology server, and it is neither clinical guidance nor a basis for clinical decisions. Nor is it complete: further work would extend coverage and put the confidence grades on firmer ground, above all through expert review.